

PACToME

Pathway ACTivation Analysis on Multiple Eukaryotes using
Reactome

User Manual

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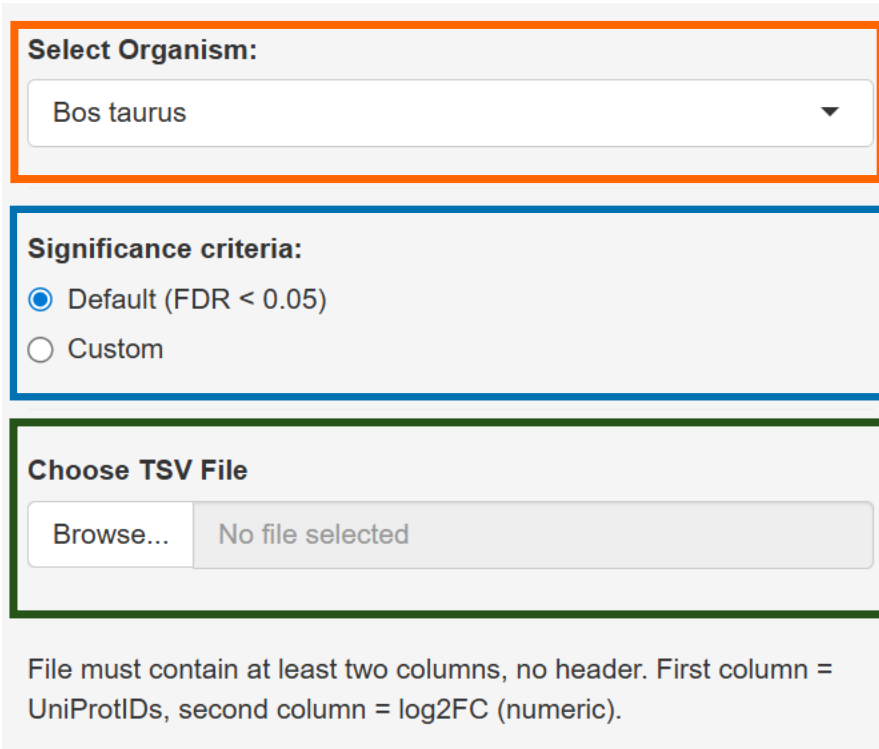
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I. Quick Start Guide

1. **Select Species** - Choose one of 14 available eukaryotic organisms from the dropdown menu. The species are listed in alphabetical order and a full list of species can be found in the PACToME manuscript.



Select Organism:

Bos taurus ▼

Significance criteria:

☒ Default (FDR < 0.05)

☐ Custom

Choose TSV File

Browse... No file selected

File must contain at least two columns, no header. First column = UniProtIDs, second column = log2FC (numeric).

2. **Set Significance Thresholds** – Default is already chosen selected or click Custom. Two input boxes will appear to type in the FDR and log₂ fold change values (can only be numeric values). This is information that the application will take into consideration to define how a pathway is deemed significant.
3. **Upload Input Data** – Upload a .tsv file with UniProt IDs in the first column and log₂ fold change values in the second column. File specifications are displayed below the file upload.
4. **Run Activation Analysis** – The analysis runs **automatically upon file upload**. Results may take up to 30 seconds to populate in corresponding tabs within the application. Ensure the appropriate species and threshold criteria are selected prior to file upload (see Application Output for more information).
5. **Download Results** – Navigate to the lower side panel to download significant and all pathway results tables (.csv) and detailed boxplots (.png).

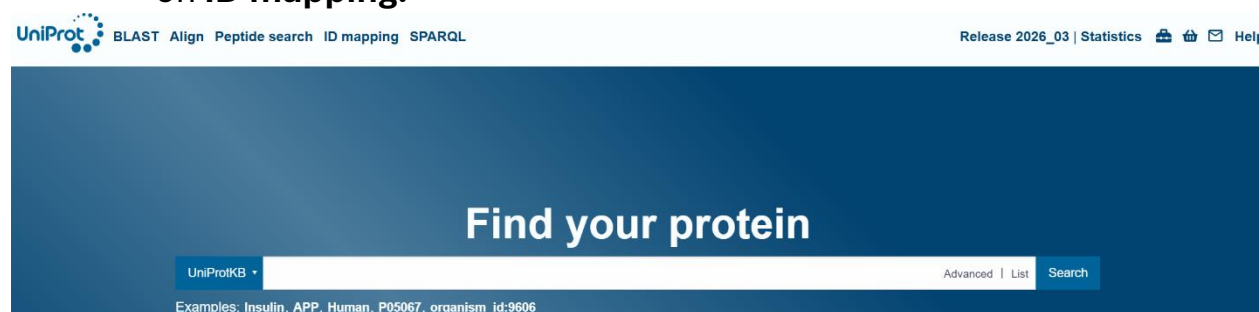
II. Input Requirements

File Format – the input file should be a tab-separated document (.tsv) containing only two columns with no headers. The first column will contain stable UniProt identifiers (IDs) and the second column will have the $\log_2$ fold change values (numeric only). A template data file is available for download in the side panel of the application and can be used to test application output.

III. UniProt ID Conversion

PACToME only accepts UniProt IDs; therefore, all gene identifiers must be converted to UniProt IDs before uploading input data to PACToME.

- A. Navigate to UniProt's website (<https://www.uniprot.org>) and click on **ID mapping**.



- B. The next screen will populate and show a text box where you can copy & paste your gene identifiers or upload a text file of identifiers by clicking on the light blue text.

Retrieve/ID mapping

Enter one or more IDs (100,000 max). You may also [load from a text file](#). Separate IDs by whitespace (space, tab, newline) or commas. This service can also be used [programmatically](#). Alternatively, the [underlying data](#) can be downloaded.

The image shows a text input field for entering UniProt IDs. The text 'P31946 P62258 ALBU_HUMAN EFTU_ECOLI' is entered into the field. The field is light blue and has a small icon in the bottom right corner.

- C. Below the text box, a drop-down menu will appear to select the database of the gene identifiers in the input file and select **UniProtKB**. In the lower left corner, select **Map IDs**.

The image shows two dropdown menus for selecting the database. The first dropdown is labeled 'From database' and has 'UniProtKB AC/ID' selected. The second dropdown is labeled 'To database' and has 'UniProtKB' selected. Both dropdowns are blue with a white arrow pointing down.

IV. Application Output

Summary Tab – the first tab located in the center of the application populates summary statistics from data in the input file. ***If the inappropriate species is chosen for analysis based on the input file's corresponding UniProt IDs, this table will populate with all zeroes.*** Below is an example of how the table populates when the analysis works. Metrics include **Total Genes in Uploaded**

Summary	Table of Sig. Pathways
Run Statistics	
Metric	Value
Total Genes in Uploaded Data	21919
Total Pathways Analyzed	1111
Pathways Meeting Criteria	66
Activated Pathways	61
Repressed Pathways	5

Data, Total Pathways Analyzed, Pathways meeting Criteria, Activated Pathways, and Repressed Pathways.

-Total Genes in Uploaded Data will correspond to the number of rows in the input data file.

-Total Pathways Analyzed will indicate the number of pathways analyzed based on the input file. More specifically, this number may differ from the total number of pathways available for a given species. For the example above, the input file

contained *C. elegans* UniProt IDs and detected 1,111 pathways and not the total number of available pathways for *C. elegans* (total = 1,126 pathways) from Reactome. The total number of available pathways depends on the version of Reactome data which will be manually updated. Please note that updates to Reactome will not automatically reflect within PACToME

-Pathways Meeting Criteria will indicate the number of significantly activated and repressed pathways based on the results from the binomial test. Criteria are the significance threshold criteria selected after species selection. The user should take into consideration the experimental design of the data for analysis or research questions to inform their decision on whether to use default or custom significance parameters.

-Activated Pathways & Repressed Pathways will showcase the split between activated and repressed pathways for the input data. The two values in these rows should always match the summation of the **Pathways Meeting Criteria** row.

IV. Application Output (continued)

Table of Sig. Pathways – the second tab located in the center of the application output populates comprehensive information on each pathway that met significance criteria. The user can choose the number of entries (pathways) they would like to view at one time in the table (default = 10; options = 10, 25, 50, 100). There is a **search** feature on the upper righthand corner of the table for the user to locate specific rows within the table. The table of significant pathways will include the following information including **Pathway ID, Pathway Name, Pathway Size, Percent of Input Genes (Pct_Input_Genes), Percentage of Activated Genes (Pct_Up), Percent of Repressed Genes (Pct_Down), PValue, and FDR.**

-**Pathway ID** is the species-specific, numerical pathway identification number in the Reactome Knowledgebase.

-**Pathway Name** is the pathway name in the Reactome Knowledgebase. This pathway name can be shared across species.

-**Pathway Size** is the total count of genes/proteins for that specific pathway based on Reactome's curation of information.

- **Percent of Input Genes (Pct_Input_Genes)** is the percentage of genes for a particular pathway detected from the user's input data. This is to give the user insight of how many genes/proteins their input data contained based on the completed curation of this pathway for a specific species. This number should be considered when interpreting the next two parameters.

- **Percentage of Activated (Pct_Up) & Repressed (Pct_Down) Genes** indicates the percentage of activated or repressed genes based on the user's input data and corresponding threshold criteria for determining pathway significance.

Boxplot of Sig. Pathways – the third tab located in the center of the application populates a boxplot of the top 10 pathways sorted by lowest False Discovery Rate (FDR). The boxplots are color coded based on the median logFC (orange = activation; blue = repression).

All Pathways Results Table – the fourth tab located in the center of the application populates all results for tested pathways.

All application output can be downloaded except the summary statistics.

V. Significance Thresholds & Customization

PACToME allows the user to choose between the default setting and

VI. Statistical Approach

PACToME utilizes a binomial test to assess whether a pathway has significantly more activated or repressed gene than expected by chance (50:50 split). The reference set of genes used for statistical analysis are all genes included in the input file. Please do not input all known genes for a species or genes that have been filtered (i.e. using log fold change as a significance threshold). P-values are FDR-corrected based on default significance or custom significance thresholds.

VI. References

Reactome Knowledgebase: <https://reactome.org/>

UniProt ID Mapping Tool: <https://www.uniprot.org/id-mapping>

Dartmouth Data Reuse and Analysis Applications: [Data Reuse and Analysis Applications – DartCF](#)

PACToME GitHub Repository: [GitHub URL]

****For complete documentation, including validation details and case studies, refer to the PACToME manuscript.****